SYNCHRONOUS — on the transaction's critical path

t = 0

50 ms SLA budget

1.37 ms consumed
(PSRG 1.2 + XGBoost 0.17, P99)

16.8 ms headroom (Δ_budget)

CDN / Edge scrubbing → API Gateway (Kong) → WAF / Service Mesh → **PSRG (Envoy WASM)** TCS-scaled rate limit 1.2 ms P99 · reads TCS cache → **Inline XGBoost** per-request score 0.170 ms P99 → Payment Service (auth / 3DS / token) → Authorization response (< 50 ms)

TCS/CLCS cache refresh, every 500 ms (gRPC) — ~~the only channel by which graph-ML~~ output reaches the SLA-bound path

ASYNCHRONOUS — off the critical path

every sync-plane component publishes here, async

5 CSDA vectors (A–E)
ACS / BIN / auth-flood / token / EDoS

FDCE decline pattern
rules R-11 / R-12 (card-fraud signal)

both attack classes enter the SAME pipeline — one architecture, not two bolted together

Kafka Event Bus
auth · 3ds · token · decline-events

Semantic Transaction Correlator (STC)
builds TIG in Neo4j
3.1 ms mean · TCS every 500ms

Baseline Integrity Monitor (BIM)
background CUSUM drift check

CLCS Engine
signal fusion, TCS-weighted
0.5 ms · outputs composite score against threshold θ_CLCS

bundle mechanisms (CDN scrub, rate-limit tightening, MTD rotation) applied at the infrastructure layer

CLCS ≥ θ_CLCS triggers pre-approved bundle activation (async, not on the current transaction's 50ms path — governs subsequent transactions)

GOVERNANCE — pre-authorized, one-time per activation

PADO
Pre-Authorized Defense Orchestrator
DORA Art. 11 · 340 ms mean, one-time

Green < 0.30 | Yellow 0.30–0.525 | Orange 0.525–0.75 | Red ≥ 0.75 (θ_CLCS)

Generic pattern (any transactional system)

Transaction state → Coherence signal
Composite score → Pre-auth. orchestration

this paper instantiates it as: ISO 8583 state machine → TCS (§IV.C) → CLCS (§IV.D) → PADO/DORA (§IV.G)